

1. Administrative & Study Identification

Full Study Title A Phase 2 Nonrandomized, Open-label, Multisite Study to Evaluate the Safety and Efficacy of Raludotatug Deruxtecan in Participants With Gastrointestinal Cancers **Short Title/Acronym** REJOICE-GI01 (MK-5909-005) **Protocol Number** MK-5909-005 **NCT Number** NCT06864169 **Sponsor Name** Merck Sharp & Dohme LLC **Investigational Product** Raludotatug Deruxtecan (R-DXd / MK-5909) **Phase of Development** Phase 2 **Medical Monitor** Medical Director, Merck Sharp & Dohme LLC

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate the Objective Response Rate (ORR) of raludotatug deruxtecan (R-DXd) to see if the cancer responds to treatment (gets smaller or goes away). * **Secondary Objectives:** To evaluate Duration of Response (DOR), Progression-Free Survival (PFS), Overall Survival (OS), and safety/tolerability of R-DXd.

2.2 Background & Rationale Researchers are looking for new ways to treat certain types of advanced gastrointestinal (GI) cancers (including pancreatic, biliary tract, colorectal, and gastroesophageal adenocarcinomas), which typically have a poor prognosis. R-DXd is an antibody-drug conjugate (ADC) designed to attach to specific proteins on cancer cells and deliver targeted cytotoxic treatment directly to those cells to destroy them.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Single-arm * **Blinding:** Open-label * **Randomization:** Nonrandomized

3.2 Planned Enrollment & Duration * **Target \$N\$:** 160 participants * **Number of Sites:** Multisite (Global, approximately 41 locations) * **Estimated Study Start:** 01-Apr-2025 * **Estimated Primary Completion:** 18-Aug-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Has one of the following unresectable or metastatic cancers: Pancreatic ductal adenocarcinoma (PDAC), adenocarcinoma of the biliary tract (CCA or GBC), colorectal adenocarcinoma, gastric adenocarcinoma, gastroesophageal junction adenocarcinoma (GEJAC), or esophageal adenocarcinoma (EAC). 2. Has received prior therapy for the cancer. 3. Has a life expectancy of at least 3 months.

4.2 Exclusion Criteria 1. History of (noninfectious) interstitial lung disease (ILD)/pneumonitis that required steroids, current ILD/pneumonitis, or suspected ILD/pneumonitis. 2. Clinically severe pulmonary compromise resulting from intercurrent pulmonary illnesses or uncontrolled/significant cardiovascular disease. 3. Active autoimmune disease that has required systemic treatment in the past 2 years, or known active central nervous system (CNS) metastases.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** Raludotatug Deruxtecan (R-DXd) administered per study protocol. * **Route of Administration:** Intravenous (IV) infusion. * **Duration of Treatment:** Continued until disease progression, unacceptable toxicity, or study completion (Est. 04-Jan-2029).

5.2 Concomitant & Prohibited Therapy * **Permitted:** Routine supportive care as defined by the protocol. * **Prohibited:** Concurrent administration of other systemic antineoplastic therapies or investigational agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Radiographic Imaging, Labs
Baseline	Day 0	Enrollment, First Dose
Interim	Ongoing Cycles	Safety Labs, Efficacy/Tumor Assessment, AE monitoring
End of Study	Follow-up	Final Vital Signs, Survival Tracking, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Objective Response Rate (ORR)
* **Measurement Method:** Assessed per RECIST v1.1 criteria.

7.2 Secondary & Exploratory Endpoints * Duration of Response (DOR) * Progression-Free Survival (PFS) (assessed by Blinded Independent Central Review - BICR) * Overall Survival (OS)

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Routine collection tracking any untoward medical occurrences temporally associated with the study intervention.
- **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness.
- **Data Monitoring Committee (DMC):** Ongoing safety evaluation monitoring.

9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) for primary efficacy; Safety Set for all AE/SAE evaluations.
- **Sample Size Justification:** N=160 to adequately evaluate ORR signals across specific GI tumor cohorts.
- **Handling of Missing Data:** Standard oncology imputation rules (e.g., participants without post-baseline scans treated as non-responders for ORR).

10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
- **Monitoring Plan:** Scheduled onsite and remote clinical monitoring visits to verify source data and safety events.
- **Audit Trail:** Comprehensive eCRF locking and query resolution forms.

11. Study Identifiers & Governance

- **Primary ID:** MK-5909-005
- **Registry Number:** NCT06864169
- **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC
- **Study Title:** A Study of Raludotatug Deruxtecan (R-DXd) in People With Gastrointestinal Cancers (MK-5909-005)
- **Official Regulatory Title:** A Phase 2 Nonrandomized, Open-label, Multisite Study to Evaluate the Safety and Efficacy of Raludotatug Deruxtecan in Participants With Gastrointestinal Cancers

12. Clinical Framework (Oncology Specific)

- **Primary Condition:** Gastrointestinal Cancers (PDAC, BTC, CRC, Gastric/GEJ/Esophageal)
- **Diagnosis Threshold:** Unresectable or metastatic disease
- **Medical Methodology:** Targeted Antibody-Drug Conjugate (ADC) therapy
- **Optimization Target:** Objective tumor response / shrinkage

13. Study Procedures & Methodology

- **Management Pathway:** Standard investigational oncology pathway for ADC administration
- **Education Protocol (HCP):** Protocol training with a strict emphasis on monitoring for ILD/pneumonitis
- **Education Protocol (Patient):** Informed consent, potential side-effect awareness (e.g., respiratory symptoms)
- **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) of imaging for unbiased efficacy evaluation

14. Observation & Follow-Up Schedule

- **Total Duration:** Until study completion (estimated Jan 2029)
- **Onsite Visit Cadence:** Aligned with treatment infusion cycles
- **Remote Monitoring Frequency:** Routine safety follow-up checks
- **Checklist Review Interval:** Radiographic tumor assessments typically performed every 6 to 8 weeks

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	_____	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	_____	_____	[DD-MMM-YYYY]					Lead
Statistician	_____	_____	[DD-MMM-YYYY]					